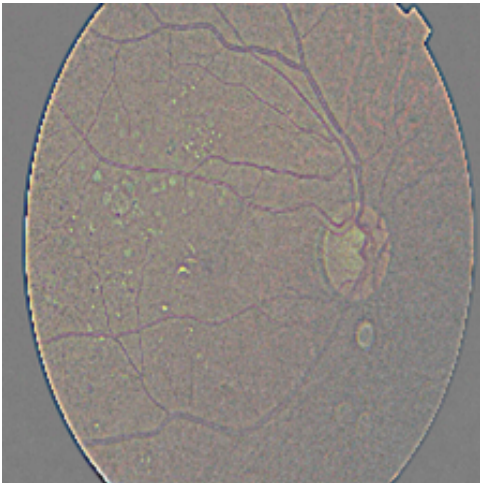


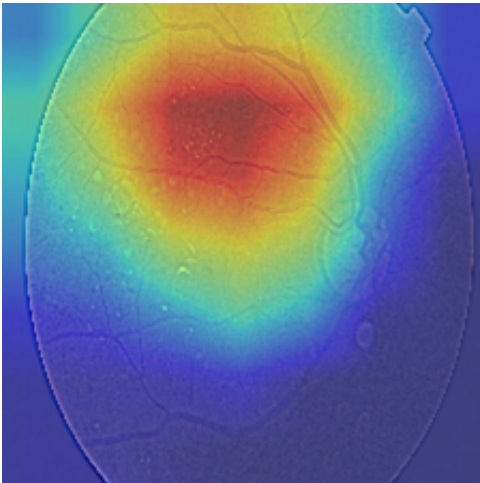
# Clinical Diagnostic & RAG Report - Diabetes

|                    |                                                                      |                     |                                    |
|--------------------|----------------------------------------------------------------------|---------------------|------------------------------------|
| Target Domain:     | Diabetes                                                             | Image File:         | Moderate_DR_867.png                |
| Predicted Class:   | Moderate DR                                                          | Confidence:         | 80.1%                              |
| 95% Conformal Set: | ['Healthy', 'Mild DR', 'Moderate DR', 'Proliferate DR', 'Severe DR'] | Uncertainty Status: | [WARNING] Human Review Recommended |

## Visual Diagnostic Imaging & Grad-CAM Heatmap



Original Medical Scan



Grad-CAM Saliency Overlay (ROI)

## AI Clinical Explanation (FLAN-T5 Grounded RAG)

'The image was classified as 'Moderate DR' with 80.1% confidence. Biomedical literature (PMID:37338737) indicates: Long-term exposure to diabetes mellitus leads to the formation of diabetic retinopathy (DR), which can cause vision loss in working-age adults. Early stage diagnosis of DR is highly essential for preventing vision loss and preserving vision in people with diabetes. The motivation behind the severity grade classification of DR is to develop an automated system that can assist ophthalmologists and healthcare professionals in the diagnosis and management of DR.'

## Faithfulness & Quantitative Text Metrics

|                         |                 |                       |        |
|-------------------------|-----------------|-----------------------|--------|
| NLI Faithfulness Score: | 0.2474 / 1.0000 | ROUGE-1:              | 0.8387 |
| BLEU-4:                 | 0.8250          | BERTScore Similarity: | 0.5730 |

## Retrieved PubMed Literature Evidence

[1] Automatic severity grade classification of diabetic retinopathy using deformable ladder Bi attention U-net and deep adaptive CNN. (PMID: 37338737)

'Long-term exposure to diabetes mellitus leads to the formation of diabetic retinopathy (DR), which can cause vision loss in working-age adults. Early stage diagnosis of DR is highly essential for preventing vision loss and preserving vision in people with diabetes. The motivation behind the severity grade classification of DR is to develop an automated system that can assist ophthalmologists and healthcare professionals in the diagnosis and management of DR.'

**[2] Simple methods for the lesion detection and severity grading of diabetic retinopathy by image processing and transfer learning. (PMID: 34488028)**

'Diabetic retinopathy (DR) has become one of the major causes of blindness. Due to the increased prevalence of diabetes worldwide, diabetic patients exhibit high probabilities of developing DR. There is a need to develop a labor-less computer-aided diagnosis system to support the clinical diagnosis.'

**[3] Advanced CNN Deep Learning Model for Diabetic Retinopathy Classification. (PMID: 42292681)**

'BACKGROUND: Diabetic Retinopathy (DR) is one of several retinal microvascular complications of Diabetes Mellitus (DM), a disease of increasing global prevalence. However, early detection and treatment can reduce or even prevent DR progression. In this work, Deep Learning (DL) techniques are used to grade DR from an early stage using either binary or multiclass classification as a clinical aid to help reduce the risk of patient vision loss.'

**[4] Retinal vascular morphological characteristics in diabetic retinopathy: an artificial intelligence study using a transfer learning system to analyze ultra-wide field images. (PMID: 38895683)**

'AIM: To investigate the morphological characteristics of retinal vessels in patients with different severity of diabetic retinopathy (DR) and in patients with or without diabetic macular edema (DME). METHODS: The 239 eyes of DR patients and 100 eyes of healthy individuals were recruited for the study. The severity of DR patients was graded as mild, moderate and severe non-proliferative diabetic retinopathy (NPDR) according to the international clinical diabetic retinopathy (ICDR) disease severity scale classification, and retinal vascular morphology was quantitatively analyzed in ultra-wide field images using RU-net and transfer learning methods.'

**[5] Automatic Detection of Diabetic Retinopathy in Retinal Fundus Photographs Based on Deep Learning Algorithm. (PMID: 31737428)**

'PURPOSE: To achieve automatic diabetic retinopathy (DR) detection in retinal fundus photographs through the use of a deep transfer learning approach using the Inception-v3 network. METHODS: A total of 19,233 eye fundus color numerical images were retrospectively obtained from 5278 adult patients presenting for DR screening. The 8816 images passed image-quality review and were graded as no apparent DR (1374 images), mild nonproliferative DR (NPDR) (2152 images), moderate NPDR (2370 images), severe NPDR (1984 images), and proliferative DR (PDR) (936 images) by eight retinal experts according to the International Clinical Diabetic Retinopathy severity scale.'